Lab 4: TokTickIT — Actions Taken, Dashboards, and Final Regression

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Peer Reviewer: Punnapob Wirojwongchai — 67070503425 — SaintCrois

Answer Part 1: Git Use with Engineering Workflow

GitHub Repository: https://github.com/Decaa0/toktickit

Every feature was isolated in a dedicated branch, peer-reviewed via pull request, approved by the peer reviewer, and merged into `lab4-staging` before final integration into `main`. Each pull request link below is the review-and-merge evidence for its issue: it shows the diff, the peer's approval, and the merge commit.

Issue	Scope	Pull Request (click to open)
#44	Issue 4.1: Sprint 4 Engineering Contract & Specifications	PR #56
#45	Issue 4.2: Actions Taken foundation (model, migration, seed, API)	PR #57
#47	Issue 4.4: Ticket workflow & resolution gate	PR #58
#48	Issue 4.5: Role dashboards (Requester & IT Staff)	PR #59
#46	Issue 4.3: Actions Taken UI	PR #60
#49	Issue 4.6: Responsive fix & final hardening	PR #61

All six pull requests were approved and merged by the peer reviewer (SaintCrois). Rendered reviewer record: `docs/lab-04/reviewer.md`. README, `.gitignore`, and the repository structure are current on `main`.

Answer Part 2: Spec DD

Rendered: `docs/lab-04/specification.md`. Includes numbered Functional Requirements (FR-01+), Business Rules (BR-01+), the complete Ticket status-transition matrix (BR-09), the IT-Staff resolution gate (BR-08), dashboard calculation rules (BR-10), two justified database-design decisions, Acceptance Criteria (AC-01+), and the Product Definition of Done. The specification was committed (PR #56) before the implementation pull requests, evidencing specification-before-code.

Answer Part 3: Test DD and Traceability

Rendered: `docs/lab-04/tests.md` — planned tests and a traceability matrix mapping each Acceptance Criterion to its automated test file. Passing suites on `main`:

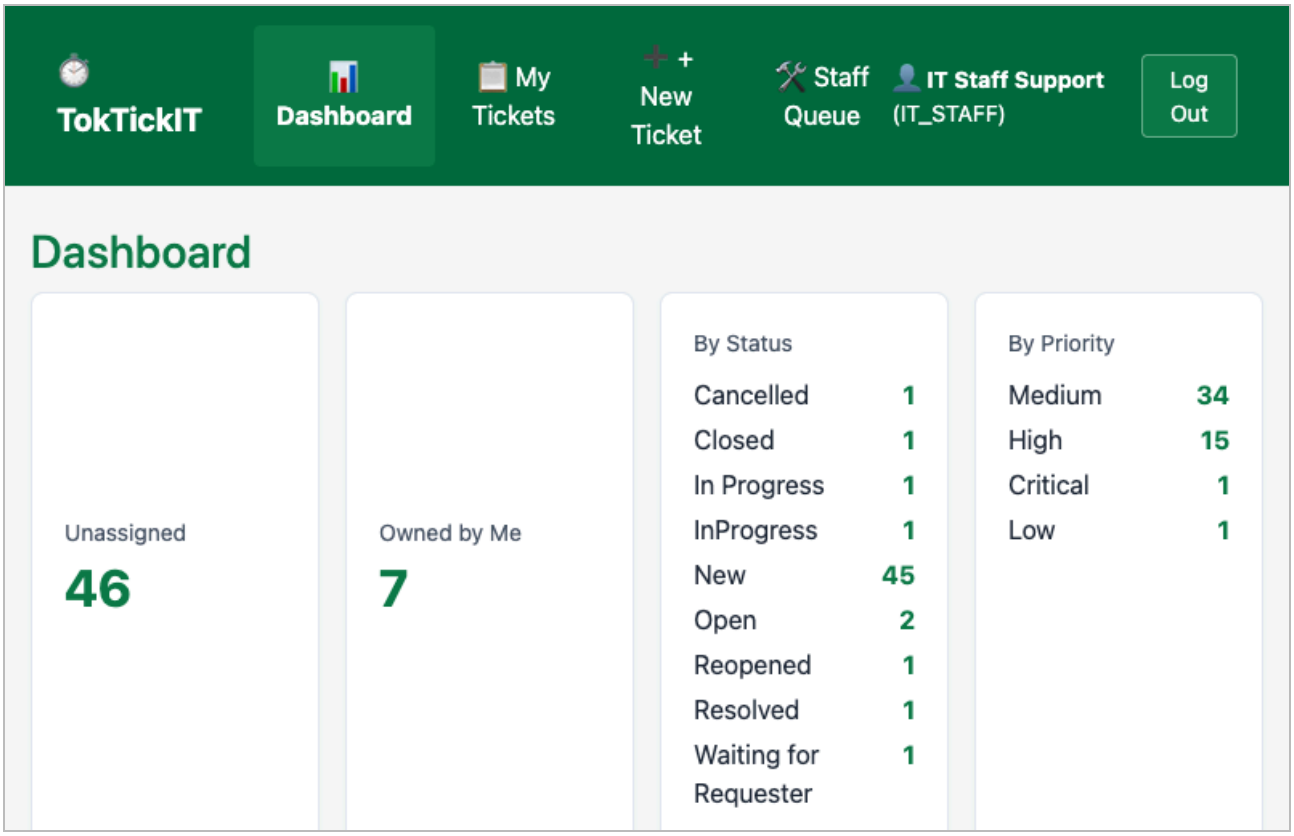
```
server:  Test Files 13 passed (13)    Tests 46 passed (46)
client:  Test Files 10 passed (10)    Tests 18 passed | 2 todo (20)
tsc --noEmit: clean        vite build: OK
```

Answer Part 4: AI Use & Reflection

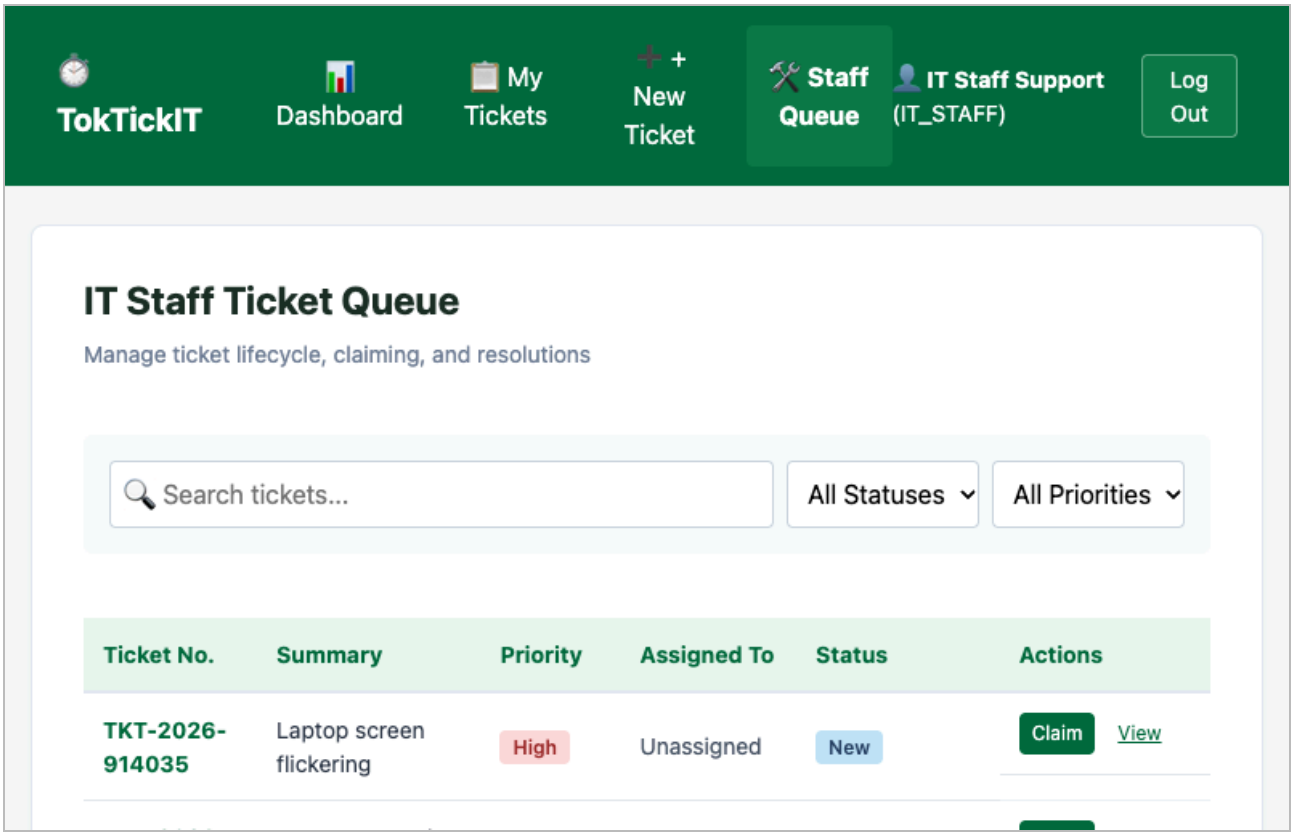
Rendered: `docs/lab-04/ai-use.md` — the LLM tools used, 8 selected key engineering prompts with their verifications, and a brief reflection on specification-agent and coding-agent use.

Answer Part 5: Working IT Staff Dashboard UI

Backend `GET /api/dashboard/staff` returns unassigned, owned-by-me, by-status, by-priority, and recently-updated metrics computed from authoritative data (verified in `staff-dashboard.api.test.ts`). Metric cards drill through to the Staff Queue.



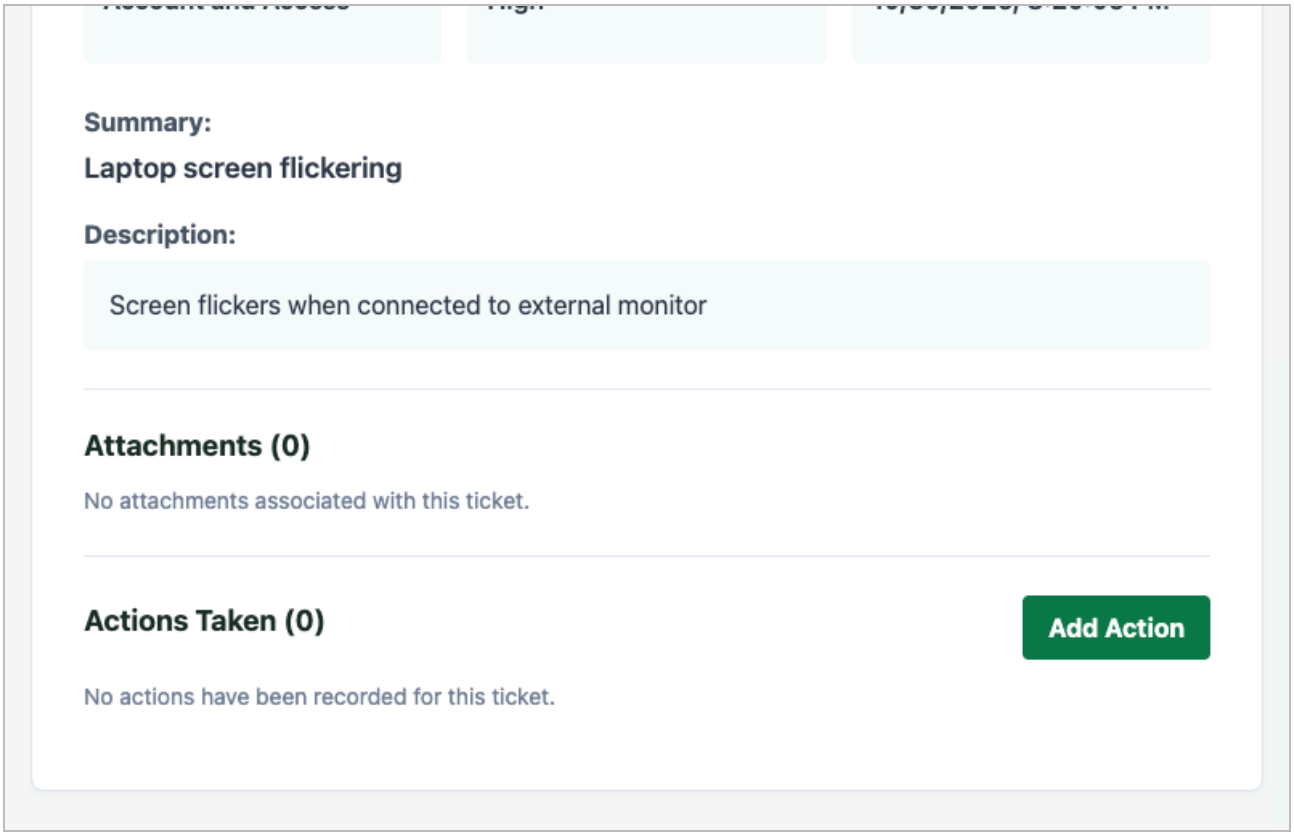
IT Staff Dashboard — operational metrics.



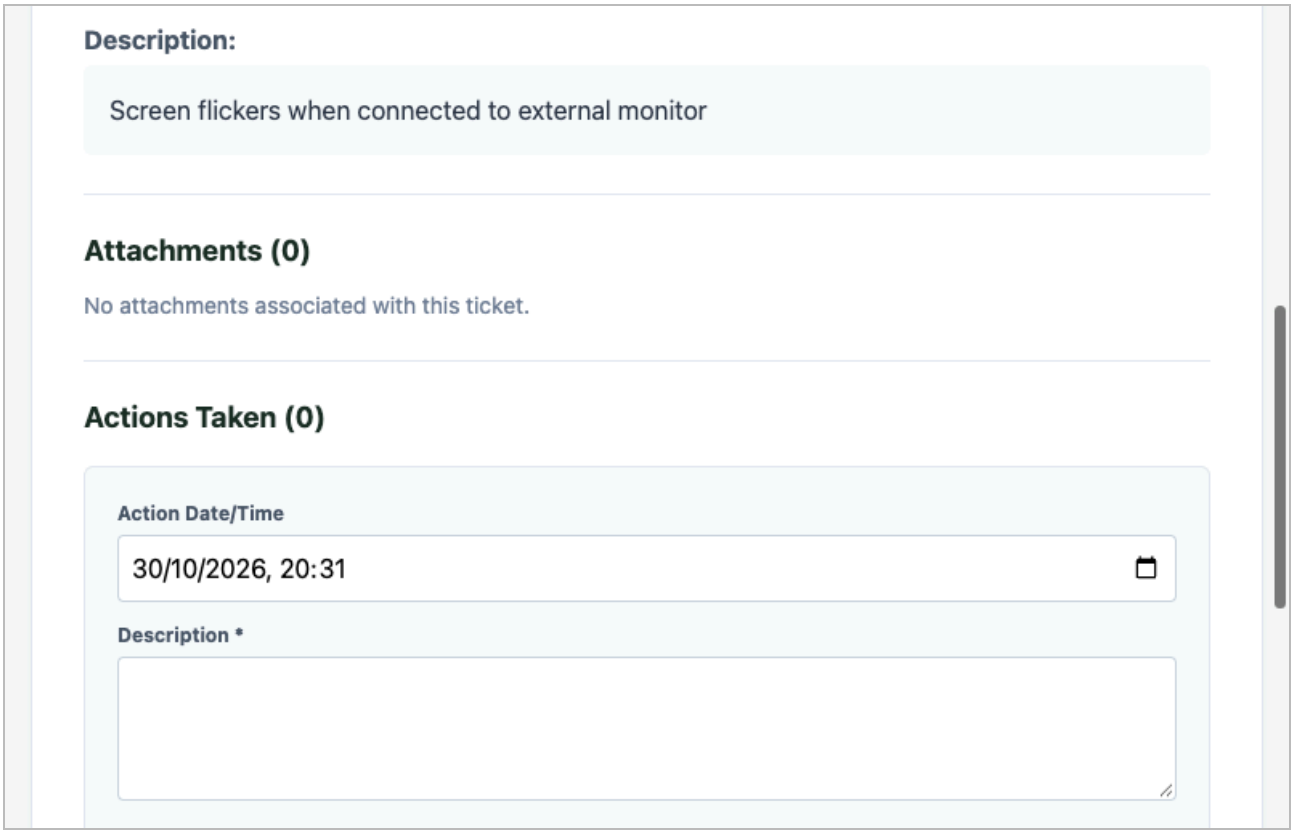
IT Staff Ticket Queue — filters, claim, and drill-down.

Answer Part 6: Working Actions Taken UI

Endpoints `GET/POST/PATCH /api/tickets/:id/actions` enforce authorization on the backend (IT Staff/Administrator write; Requester reads only owned tickets) and require a follow-up note when follow-up is flagged. The UI supports a list, create and edit modes, a duplicate-submit guard, and read-only display for Requesters.



Ticket Detail with the Actions Taken section and "Add Action".



Actions Taken create form (Action Date/Time, Description, follow-up fields).

Answer Part 7: Working Ticket Workflow

The backend enforces the full eight-status transition matrix (New, Open, In Progress, Waiting for Requester, Resolved, Closed, Reopened, Cancelled). Illegal transitions return 400; stale updates return 409; a Requester "appears resolved" indication is advisory and never sets the ticket to Resolved. Verified in `ticket-workflow.api.test.ts`. Status controls in the Ticket Queue (above) expose only permitted transitions.

Code capture — the authoritative transition matrix (`server/src/controllers/staff.controller.ts`):

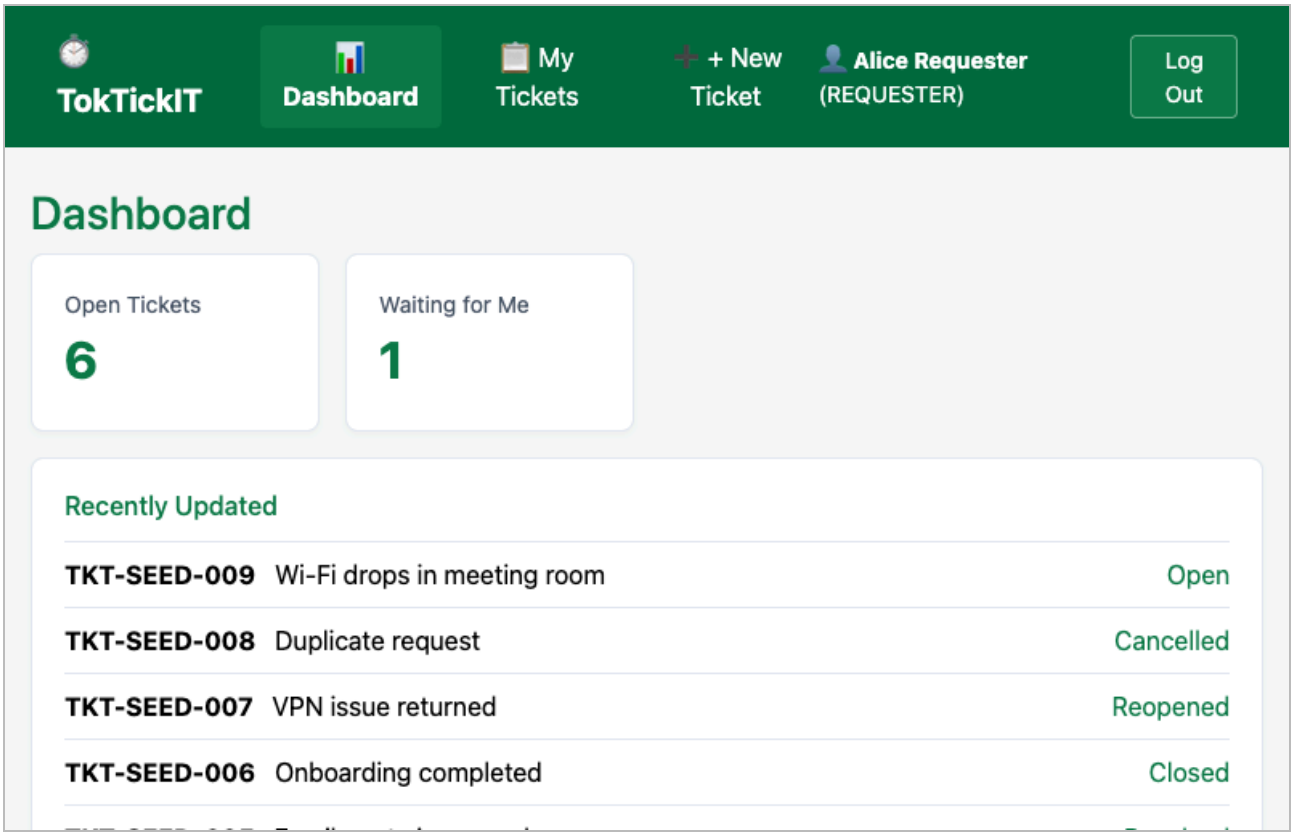
```
const VALID_TRANSITIONS: Record<string, string[]> = {
  "New": ["Open", "Cancelled"],
  "Open": ["In Progress", "Waiting for Requester", "Resolved", "Cancelled"],
  "In Progress": ["Waiting for Requester", "Resolved", "Cancelled"],
  "Waiting for Requester": ["In Progress", "Resolved", "Cancelled"],
  "Resolved": ["Closed", "Reopened"],
  "Reopened": ["In Progress", "Waiting for Requester", "Resolved",
"Cancelled"],
  "Closed": ["Reopened"],
  "Cancelled": [],
};
```

Code capture — optimistic-concurrency guard returning HTTP 409 on a stale update:

```
if (expectedUpdatedAt && ticket.updatedAt > new Date(expectedUpdatedAt)) {
  return res.status(409).json({
    error: "Conflict: ticket was modified by someone else",
    currentUpdatedAt: ticket.updatedAt,
  });
}
```

Answer Part 8: Working Requester Dashboard and Final Regression

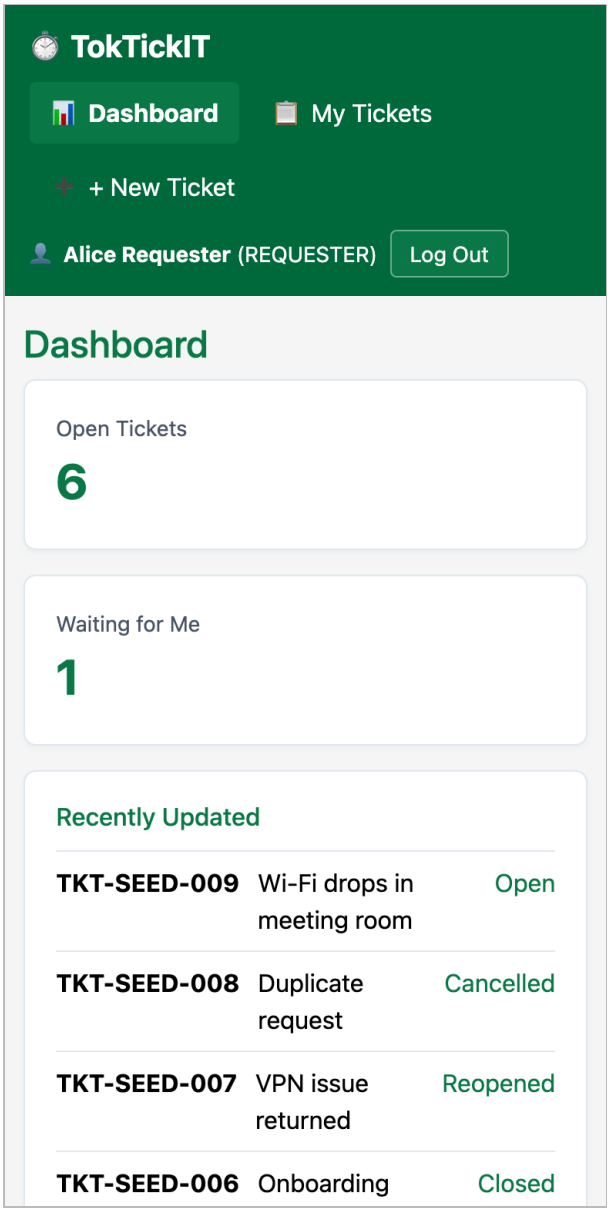
The Requester dashboard (`GET /api/dashboard/requester`) is owner-scoped: total open, waiting-for-me, recently updated, and recently resolved — ownership enforced on the backend (verified in `requester-dashboard.api.test.ts`). All Lab 1–3 suites continue to pass on `main`, confirming preserved authentication, My Tickets, Ticket Detail, attachments, IT Staff functions, and administrator user management.



Requester Dashboard — owner-scoped metrics and recent tickets.

Answer Part 9: Zen Green UI and Responsive Evidence

Rendered: `docs/lab-04/ui-spec.md`. Root-cause fix for the mobile defect: the application previously served a stale compiled `App.js`, so edits to `App.tsx` never shipped; `main.tsx` and the tests now load `App.tsx`, and `tsconfig noEmit:true` prevents re-emitting the compiled files. Responsive styles add breakpoints at 768px and 1024px, a wrapping navigation bar, stacked cards, full-width inputs (16px to prevent iOS zoom), and visible focus outlines; no horizontal page scroll remains.



Mobile (390px) view — nav wraps, cards stack, no horizontal scroll.

The submitted repository and final `main` branch remain the source of truth.